

Machine Learning-Driven Materials Design for Additive Manufacturing

Project Requirements and Specifications

Dr. Arezoo Zare



Albert Li

Anh Duong

Ryan Calimlim

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1. Introduction

For this Capstone Project we will be working with the Washington State University School of Mechanical and Materials Engineering. We will be building off a previous team's capstone project, fixing bugs, and expanding the capabilities already implemented. The existing project is a desktop program which accepts microscopy images of various materials and then identifies different inclusions in the material. These inclusions are normally a secondary phase

of the surrounding compound or a different compound altogether, the addition of which changes the characteristics of the material.

In addition to the bug fixing, we plan on incorporating a forward model that will allow a user to input a microscopy image of a material, and the program will output a preliminary assessment of that material's physical characteristics. Should time permit, we also hope to implement an inverse model that will allow a user to input the desired physical characteristics of a material, and the program will specify the type and composition of the inclusions that will result in said characteristics.

2. System Requirements Specification

2.1. Use Cases

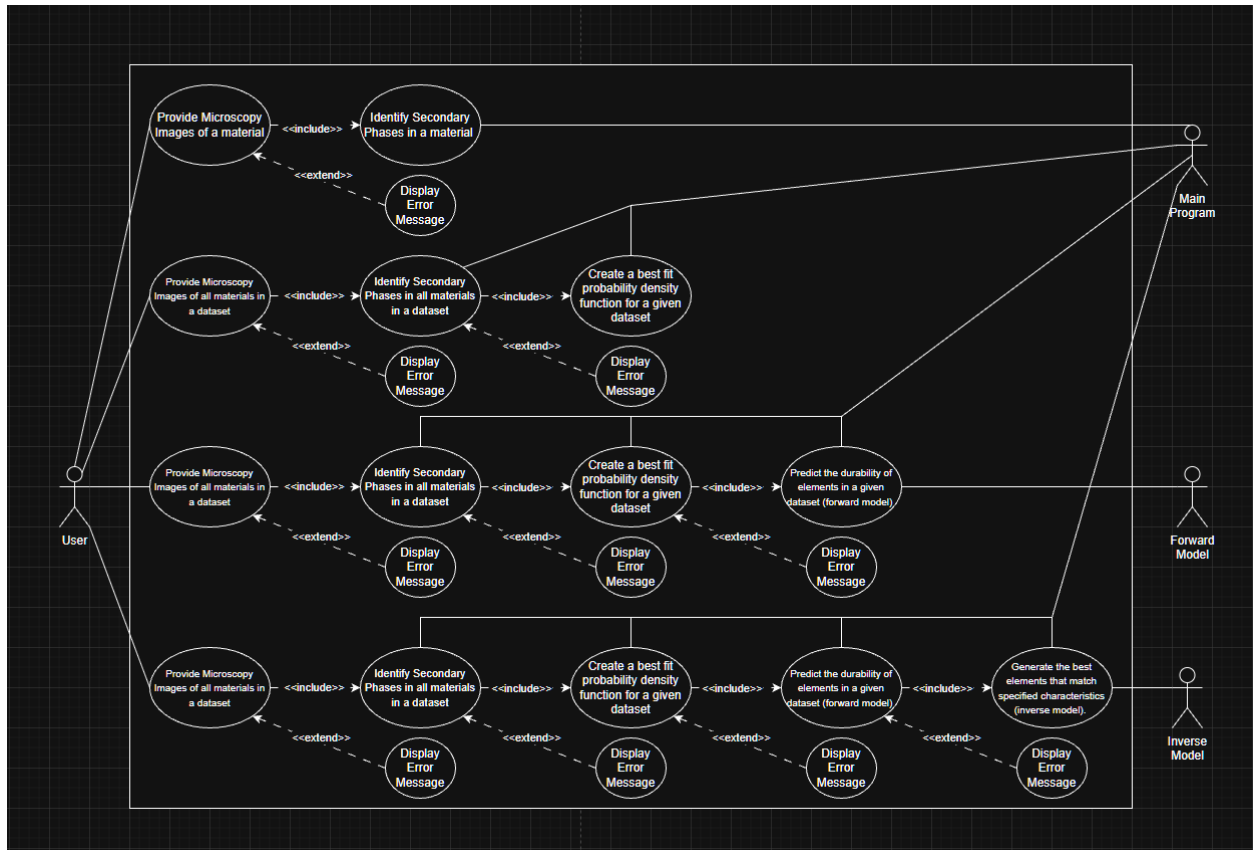


Figure 1: Use Case Diagram

Use Case	Identify Secondary Phases in a material
Actors	Researcher
Pre-Condition	-Microscopy Images of the material are provided.
Post-Condition	-Secondary phases in the material are identified and located.
Main Flow	-User inputs microscopy images of the material -User runs the program. -The secondary phases are identified and located
Alternative Flow	-Program fails -> send error message.

Related Requirements	-Functional requirements: • FR-2 • FR-3 • FR-4
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Use Case	Create a best fit probability density function for a given dataset
Actors	Researcher
Pre-Condition	-Processed microscopy images of the materials in the given dataset are provided (secondary phases have been identified and located by a previous step of the program).
Post-Condition	-A best fit probability density function is created for the given dataset.
Main Flow	-User inputs processed microscopic images of the materials in the dataset. -User runs the program. -A best fit probability density function is created for the given dataset.
Alternative Flow	-Program fails -> send error message.
Related Requirements	-Functional Requirements: • FR-1

Use Case	Predict the durability of elements in a given dataset (forward model)
Actors	Researcher
Pre-Condition	-Processed microscopy images of the materials in the given dataset are provided (secondary phases have been identified and located by a previous step of the program). -A best fit probability density function for the given dataset is provided (the function has been created by a previous step of the program).
Post-Condition	-The elements in the given dataset have their durability predicted (a forward model is created).
Main Flow	-User inputs processed microscopic images of the materials in the dataset. -User inputs the best fit probability density function created for the given dataset. -User runs the program. -A machine learning model attempts to predict the durability of elements in the given dataset. -The durability prediction (forward model) is created for the given dataset.
Alternative Flow	-Program fails -> send error message.
Related Requirements	-Functional Requirements: • FR-5

Use Case	Generate the best elements that match specified characteristics (inverse model).
Actors	Researcher
Pre-Condition	-User inputs desired material characteristics. -User inputs the best fit probability density function created for the given material/dataset. -User inputs the forward model for the given material/dataset.
Post-Condition	-An inverse model specifying the best elements to match the desired material characteristics is created.
Main Flow	-User inputs desired material characteristics. -User inputs the best fit probability density function created for the given material/dataset. -User inputs the forward model for the given material/dataset -User runs the program. -A machine learning model attempts to predict the best elements to use that match the specified characteristics. -The element composition prediction (inverse model) is created for the desired characteristics.
Alternative Flow	-Program fails -> send error message.
Related Requirements	-Functional Requirements: • FR-6

2.2. User Stories

User Story US1: Identify Secondary Phase(s) in a material

As a user, I want to identify secondary phase(s) in the material I provide.

Feature: Identify Secondary Phase(s) in a material

Scenario: User wants to identify secondary phase(s) in a material

Given the user is on the main program page

When they enter microscopy images of their material

And select the Identify Secondary Phase(s) option

Then the program should identify and locate the different secondary phase(s) present in said material

User Story US2: Create a best fit probability density function for a given dataset

As a user, I want to create a best fit probability density function for a given dataset I provide.

Feature: Create a best fit probability density function for a given dataset

Scenario: User wants to create a best fit probability density function for a given dataset

Given the user is on the main program page

When they enter processed microscopy images of the materials in their dataset

And select the create a best fit probability density function option

Then the program should create a best fit probability density function for the given dataset

User Story US3: Predict the durability of elements in a given dataset (forward model)

As a user, I want to predict the durability of elements in a dataset I provide.

Feature: Create a forward model to predict the durability of elements for a given dataset

Scenario: User wants to create a forward model to predict the durability of elements for a given dataset

Given the user is on the main program page

When they enter processed microscopy images of the materials in their dataset

And enter the best fit probability density function created for the given dataset

And select the create a forward model option

Then the program should create a forward model to predict the durability of elements for a given dataset

User Story US4: Generate the best elements that match specified characteristics (inverse model)

As a user, I want to generate the best elements that match specified characteristics I provide.

Feature: Create an inverse model to predict the best elements for a given set of characteristics

Scenario: User wants to create an inverse model to predict the elements that will create a given set of characteristics

Given the user is on the main program page

When the user enters the desired material characteristics.

And enters the best fit probability density function created for the given material/dataset

And enters the forward model for the given material/dataset

And select the create an inverse model option

Then the program should create an inverse model to predict the elements that will create a given set of characteristics

2.3. Traceability Matrix

Functional Requirement	Use Case	User Story	Priority
[FR-1] Best fit probability density function	UC2: Create a best fit probability density function for a given dataset	US2: As a user, I want to create a best fit probability density function for a given dataset I provide.	Level 0
[FR-2] Image processing	UC1: Identify Secondary Phases in a material	US1: As a user, I want to identify secondary phase(s) in the material I provide.	Level 0
[FR-3] User image upload	UC1: Identify Secondary Phases in a material	US1: As a user, I want to identify secondary phase(s) in the material I provide.	Level 0
[FR-4] User segmentation options	UC1: Identify Secondary Phases in a material	US1: As a user, I want to identify secondary phase(s) in the material I provide.	Level 0
[FR-5] Forward model	UC3: Predict the durability of elements in a given dataset (forward model)	US3: As a user, I want to predict the durability of elements in a dataset I provide.	Level 0
[FR-6] Inverse model	UC4: Generate the best elements that match specified characteristics (inverse model)	US4: As a user, I want to generate the best elements that match specified characteristics I provide.	Level 1

Functional Requirements

2.3.1. Image segmentation

Functional Requirement	[FR-1] Best fit probability density function
Description	The system must generate a consistent and accurate best fit probability density function for any given dataset
Source	Client
Priority	0 (essential)

Functional Requirement	[FR-2] Image processing
Description	The system must use a machine learning model to analyze images in any given dataset
Source	Client
Priority	0 (essential)

Functional Requirement	[FR-3] User image upload
Description	Users must be able to upload an image or a dataset of images to be cropped and segmented by the machine learning model

Source	Client
Priority	0 (essential)

Functional Requirement	[FR-4] User segmentation options
Description	The user must be able to choose from a variety of options when performing segmentation, such as changing blur type and smooth kernel size
Source	Client
Priority	0 (essential)

2.3.2. Dataset Machine Learning Modeling

Functional Requirement	[FR-5] Forward model
Description	The system must use a machine learning model to predict the durability of elements in a given dataset
Source	Client
Priority	0 (essential)

Functional Requirement	[FR-6] Inverse model
Description	The system must use a machine learning model and the forward model to generate an inverse model that helps predict the best elements to use that would match specified characteristics
Source	Client
Priority	1 (Desirable)

2.4. Non-Functional Requirements

NFR-1 Performance: The system shall process microscopy images, generate probability density functions, and execute forward and inverse models within a reasonable time suitable for research use. Client 0 (Essential)
NFR-2 Reliability: The system shall operate consistently when processing datasets and shall provide clear error messages when invalid or incompatible inputs are provided. Client 0 (Essential)
NFR-3 Usability: The system shall provide an intuitive desktop interface that allows researchers to upload images, execute segmentation, and run predictive models without requiring programming knowledge. Client 0 (Essential)
NFR-4 Maintainability: The system shall be structured in a modular way to allow future capstone teams to fix bugs and expand system capabilities. Client 1 (Important)
NFR-5 Compatibility: The system shall function as a desktop application compatible with laboratory computing environments. Client 1 (Important)

NFR-6 Data Integrity: The system shall preserve uploaded datasets and generated model outputs during active user sessions. Client 0 (Essential)

2.5. Standards

Standard/Code	Domain	Description	Application
IEEE 830-1998 – Software Requirements Specification	Software Documentation	Defines the structure and content of software requirements specifications (SRS)	Adopted to organize and verify functional and non-functional requirements, use cases, and the traceability matrix
IEEE 1016-2009 – Software Design Description	Design Documentation	Standardizes the representation of software architecture, interfaces, and modules	Applied in diagrams for system architecture documentation
ACM Code of Ethics and Professional Conduct (2018)	Ethics/Professional Responsibility	Outlines principles of fairness, honesty, and respect in computing professions	Guided ethical AI design to ensure transparency and bias mitigation

3. System Evolution

We are working on top of code built from a previous capstone using Python which runs correctly on existing data, albeit with some UI and computational inaccuracies. Currently, some issues exist regarding generating consistent probability density functions for existing datasets, and additionally, the program fails to generate correct results for new datasets. This is likely related to the segmentation process. This previous code uses the Segment Anything Model (SAM) by Meta to segment the images, but if it becomes a problem and the segmentation algorithm is uncorrectable, pivoting to another model should be simple. For example, there are newer forks of the SAM, including HQ-SAM (High Quality – Segment Anything Model), which improves mask quality and boundary precision over the SAM, or YOLO11, an alternative computer vision model.

Our main feature to implement here is the forward model, using machine learning to predict durability, by outputting a preliminary assessment of any inputted material. Since no model currently exists for this task, and training a new model would be much too costly, we plan to use Low-Ranking Adaptation, a technique to drastically reduce computational costs of training Large Language Models. When creating the inverse model, if inverting the forward model becomes an issue, a possibility would be to switch to a mixture density network, which excels at inverse problems.

4. Glossary

Define technical terms used in the document.

Probability density function – Mathematical function that describes the relative likelihood of a continuous random variable taking on a specific value.

Microscopy image – A photograph captured through a microscope, revealing objects too small to be seen by the naked eye.

Forward model – A computational mechanism that predicts the future state of a system based on its current state and a control signal.

Inverse model – A computational mechanism that infers unknown parameters from observed data.

5. References

^[1] https://en.wikipedia.org/wiki/Use_case

^[2] https://en.wikipedia.org/wiki/Functional_requirement

^[3] https://en.wikipedia.org/wiki/Non-functional_requirement